

Peer Review: When Blocks Build Societies Multi-Agent Systems in Minecraft

Thank you for the opportunity to review the draft of your seminar paper. First of all, your paper is already very extensive, especially for this early stage of the seminar course. Moreover, it shows a clear structure, which is well suited to the scope of your research.

In addition to the general structure of your paper, the comparative analysis is well organized. The division into separate subsections for each comparison criterion made it easy to follow the logical path of your comparison. Moreover, both case studies were effectively contrasted at each stage of the analysis, with first implications at the end of each criterion stage. This helps to understand the line of argument and the derived implications.

Furthermore, especially notable, is the declaration of your generative artificial intelligence (AI) usage. Your paper provides clear transparency by stating the exact application cases of AI. Additionally, the inclusion of your complete record of interactions in the appendix, ensures full transparency to the reader. Next to providing information about AI usage, you also implemented a verification process, addressing a key aspect when working with AI, which is very important.

Major Comments

Your research is very ambitious, with clearly defined objectives and methodology. However, I would suggest to narrow or alter the scope of your paper. This is due to multiple reasons.

First, your objective aims to identify generalized (not Minecraft related) design principles for collaborative AI systems. This is coherent with the implications you provide in the comparative analysis and the discussion. However, your study is only based on two case studies which are limited to a specific application domain (Minecraft). Deriving generalized design principles for collaborative AI systems from such a small and narrow focused sample, might not be sufficiently founded. To address this methodological weakness, I would suggest to increase the sample size to a sufficiently high number and across different domains, to draw generalized implications. Alternatively, I would suggest to simply adapt the scope.

Moreover, your research does not define a clear research problem, it aims to address. Within your introduction you motivate the topic of multi agent systems, as well as Minecraft's research potential. Further, you also mention the gap between narrowly defined task oriented systems and the more complex socially oriented agent societies. However, your research does not address this specified gap. Your research contrasts two approaches which address different aspects of this gap and aims to identify design principles for collaborative AI systems. Therefore, I would suggest to clearly define a research problem that matches your objectives and your research question. This problem could for example motivate, why design principles for collaborative AI systems are needed or why it is important to contrast the two considered Minecraft use cases.

Finally, in chapter 7.2 (*Practical Applications*), you transfer the insights gained from collaborative AI systems in the Minecraft domain to other application fields like warehouse robotics, manufacturing systems, disaster response and community management in online problems. This transfer is creative and well thought out, but the reasoning is hard to understand given the lack of description for each transfer. The transfer would be more transparent if you state and explain clear aspects and parallels of each application area and really explain why the aspects you observed in Minecraft case studies can be transferred to the targeted domain. Moreover, this transfer to other domains is not listed within the scope or objectives of your research. Given that

this transfer takes place in the discussion of your paper, I would suggest adding it to the scope of your research.

Minor Comments

- During the course of your paper numerous central and especially non trivial points are not backed by any source. These need to be addressed.
- *Methodology:* You justify the selection of your two case studies based on their status as state of the art examples and their clearly contrasting approaches. However, this does not explain why your research does not consider other modern approaches. The consideration of additional modern approaches could potentially enrich your comparison and could help to establish a better fundamental base for generalized implications.
 - This selection is especially hard to follow since your paper cites Chen et al. (2024), another modern multi agent system Minecraft use case, in the comparative analysis, the discussion as well as in the conclusion. However, this approach is not considered as an actual use case in your paper. The reasoning for this is not transparent to the reader.
 - I would suggest adding an explanation to the methodology which further states and justifies the inclusion and exclusion criteria for your paper's case studies.
- *Methodology:* In addition to your already very well defined GenAI usage chapter, you could further mention the specific LLM you utilized to achieve even more transparency.
- *Methodology:* You utilized two non peer reviewed papers. This is fine, especially since your objective was to consider state of the art approaches. However, I would suggest to integrate this explanation into the methodology. This way, the reader is informed that the considered approaches are not peer reviewed, while also understanding the explanation for including non peer reviewed sources.
- *Background:* Agents and AI are central components of your research. However, both terms are not uniformly defined within literature. For clarity reasons I would suggest to define both terms in the background.
- Lastly, I would suggest to utilize the given template and to shorten the pure textual content to the limit of 15 pages.